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Air conditioner

Abstract

An air conditioner includes a housing comprising an outlet, a panel attachable to or detachable from a lower side of the housing, the panel comprising an inlet to introduce air therethrough, a first filter installable to the panel to filter the introduced air, a second filter installable to the panel at an upper side of the first filter and configured to further filter the introduced air, and a lifter configured to detach the panel from the lower side of the housing by lowering the panel from an initial position to a maintenance position in a vertical direction, and to attach the panel to the lower side of the housing by lifting the panel from the maintenance position to the initial position in the vertical direction. By lowering the panel to the maintenance position, one or more of the first filter and the second filter are accessible for the maintenance.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Ser. No. 17/007,627, filed Aug. 31, 2020, and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2019-0107583, filed on Aug. 30, 2019, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(1) The disclosure relates to an air conditioner, and more particularly, to an air conditioner including an improved lifting and lowering structure of a panel.

2. Description of the Related Art

(2) In general, an air conditioner is a household appliance that maintains indoor air at a comfortable temperature by using a refrigeration cycle of a refrigerant. The air conditioner is composed of an indoor unit that is placed in the indoor space and provided with a heat exchanger and a blower fan, an outdoor unit that is placed in the outdoor space and provided with a compressor and a condenser, and a refrigerant pipe that connects the indoor unit to the outdoor unit and circulates a refrigerant.

(3) Recently, an air conditioner, in which a dust collecting filter is additionally installed due to difficulties such as fine dust, so as to maintain a comfortable temperature in the indoor space while purifying indoor air, has been disclosed.

(4) According to the installation location of the indoor unit, the air conditioner may be classified into a stand-type air conditioner in which the indoor unit is placed on a floor, a wall-mount type air conditioner in which the indoor unit is installed on a wall, and a ceiling-type air conditioner in which the indoor unit is installed on a ceiling. The ceiling-type air conditioner is provided to be embedded in the ceiling or suspended from the ceiling.

(5) It is difficult to perform maintenance on a filter disposed in the inside of the indoor unit, because the indoor unit of the ceiling-type air conditioner is installed on the ceiling.

SUMMARY

(6) Therefore, it is an aspect of the disclosure to provide an air conditioner capable of allowing a panel and a filter of an indoor unit to be lifted and lowered so as to easily perform maintenance on the filter.

(7) It is another aspect of the disclosure to provide an air conditioner capable of allowing an electrostatic precipitator filter, which is configured to collect foreign substances using electricity, to be lifted and lowered along with a panel.

(8) It is another aspect of the disclosure to provide an air conditioner capable of easily supplying electricity to an electrostatic precipitator filter, which is configured to be lifted and lowered, when the electrostatic precipitator filter is inserted into a housing.

(9) Additional aspects of the disclosure will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the disclosure.

(10) In accordance with an aspect of the disclosure, an air conditioner includes a housing including an outlet, a panel attachable to or detachable from a lower side of the housing and including an inlet to introduce air therethrough, a first filter installable to the panel, a second filter installable to the panel to filter the introduced air and configured to further filter the introduced air using an electrostatic force, and a lifter configured to detach the panel from the lower side of the housing by lowering the panel from an initial position, and to attach the panel to the lower side of the housing by lifting the panel from a maintenance position. When the panel is lowered to the maintenance position, one or more of the first filter and the second filter are accessible for maintenance thereon.

(11) The second filter is an electrostatic precipitator filter and the air conditioner. The air

conditioner may further include a power supply disposed inside of the housing and configured to supply power to the second filter to generate the electrostatic force to filter the introduced air.

(12) The panel, the first filter and the second filter installed in the panel may be lifted or lowered in a vertical direction while maintaining a horizontal state

(13) The second filter may include a contact portion configured to be in contact with the power supply in an up and down direction.

(14) The contact portion may be disposed on an upper portion of the second filter.

(15) The power supply may include an intermediate member configured to be in contact with the contact portion electrically by being pressed by the contact portion when the panel is lifted from the maintenance position.

(16) The intermediate member may be a plate spring member.

(17) The panel, the first filter and the second filter may be lifted or lowered in the vertical direction while maintaining the horizontal state.

(18) The housing further may include a guide configured to guide a lifting direction of the panel to allow the contact portion to be in contact with the power supply in the up and down direction.

(19) The guide may include a guide surface configured to guide the lifting direction of the panel by contacting with at least a part of the panel upon lifting of the panel.

(20) The guide may include a guide rib configured to guide the lifting direction of the panel by contacting with at least a part of the panel upon lifting of the panel.

(21) The panel may further include a frame configured to support the first filter and the second filter against the panel.

(22) The second filter may be installed to the panel in a first direction corresponding to a lifting and lowering direction of the second filter, and the first filter may be installed to the panel in a second direction perpendicular to the first direction.

(23) The lifter may include four motors, and four wires connected to the four motors and connected to the panel to lift or lower the panel by driving force of the four motors.

(24) The lifter may include a motor, and a wire including one end connected to the motor and an other end connected to the panel, the wire configured to lift or lower the panel by driving force of the motor, and the panel may include an elastic member including one end connected to an other end of the wire and the other end connected to the panel.

(25) The panel may further include a pulley through which the wire passes, and the other end of the wire may pass through the pulley and then be arranged perpendicular to the one end of the wire.

(26) In accordance with another aspect of the disclosure, an air conditioner includes a housing including an outlet, a panel assembly attachable to or detachable from a lower side of the housing and configured to be moved in an up and down direction with respect to the housing, and a lifter disposed inside of the housing and configured to lift or lower the panel assembly in a vertical direction while maintaining a horizontal state of the panel assembly. The panel assembly includes a panel including an inlet to introduce air therethrough and configured to cover the lower side of the housing, a first filter installable to the panel and configured to filter the introduced air, and a second filter installable to the panel and configured to further filter the introduced air using an electrostatic force.

(27) The second filter is an electrostatic precipitator filter and the air conditioner. The air conditioner may further include a power supply disposed inside of the housing and configured to supply power to the second filter to generate the electrostatic force to filter the introduced air, and the second filter may include a contact portion configured to be in contact with the power supply in an up and down direction.

(28) The contact portion may be disposed on an upper portion of the second filter.

(29) The lifter may include four motors, and four wires connected to the four motors and connected to the panel to lift or lower the panel by driving force of the four motors.

(30) In accordance with another aspect of the disclosure, an air conditioner includes a housing

including an outlet, a panel attachable to or detachable from a lower side of the housing and including an inlet to introduce air therethrough, a first filter installable to the panel, a second filter installable to the panel and configured to filter out the introduced air using an electrostatic force, and a lifter configured to detach the panel from the lower side of the housing by lowering the panel from an initial position, and to attach the panel to the lower side of the housing by lifting the panel from a maintenance. The second filter includes a contact portion disposed on an upper portion of the second filter and configured to receive power therethrough.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and/or other aspects of the disclosure will become apparent and more readily appreciated from the following description of embodiments, taken in conjunction with the accompanying drawings of which:
- (2) FIG. 1 is a perspective view of an air conditioner according to an embodiment of the disclosure;
- (3) FIG. 2 is a side sectional view of the air conditioner according to an embodiment of the disclosure;
- (4) FIG. 3 is a view illustrating a state in which a panel assembly of the air conditioner of FIG. 1 is lowered;
- (5) FIG. 4 is a view illustrating a part of configuration of the air conditioner of FIG. 3;
- (6) FIG. 5 is a bottom view of a housing of the air conditioner according to an embodiment of the disclosure;
- (7) FIG. 6 is an exploded perspective view of the panel assembly of the air conditioner according to an embodiment of the disclosure;
- (8) FIG. 7 is a cross-sectional perspective view illustrating a part of configuration when the panel assembly of the air conditioner according to an embodiment of the disclosure is lifted;
- (9) FIG. 8 is a cross-sectional view illustrating a part of configuration when the panel assembly of the air conditioner according to an embodiment of the disclosure is lifted;
- (10) FIG. 9 is an enlarged view illustrating a part of configuration of the panel assembly of the air conditioner according to an embodiment of the disclosure;
- (11) FIG. 10 is a cross-sectional perspective view illustrating a part of configuration when a panel assembly of an air conditioner according to another embodiment of the disclosure is lifted;
- (12) FIG. 11 is a bottom view illustrating a support panel of a housing of an air conditioner according to still another embodiment of the disclosure;
- (13) FIG. 12 is a cross-sectional view illustrating a part of configuration when a panel assembly of the air conditioner according to still another embodiment of the disclosure is lifted; and
- (14) FIG. 13 is a cross-sectional view illustrating a part of configuration when the panel assembly of the air conditioner according to still another embodiment of the disclosure is lifted.

DETAILED DESCRIPTION

- (15) Embodiments described in the disclosure and configurations illustrated in the drawings are merely examples of the embodiments of the disclosure, and may be modified in various different ways at the time of filing of the present application to replace the embodiments and drawings of the disclosure.
- (16) In addition, the same reference numerals or signs illustrated in the drawings of the disclosure indicate elements or components performing substantially the same function.
- (17) Also, the terms used herein are used to describe the embodiments and are not intended to limit and/or restrict the disclosure. The singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. In this disclosure, the terms “including”, “having”, and the like are used to specify features, numbers, steps, operations,

elements, components, or combinations thereof, but do not preclude the presence or addition of one or more of the features, elements, steps, operations, elements, components, or combinations thereof.

(18) The term of “and/or” includes a plurality of combinations of relevant items or any one item among a plurality of relevant items.

(19) In the following detailed description, the terms of “upper portion”, “lower portion”, “lower side”, “front end” and “front, rear, left and right side” may be defined by the drawings, but the shape and the location of the component is not limited by the term.

(20) A refrigeration cycle forming an air conditioner is composed of a compressor, a condenser, an expansion valve, and an evaporator. The refrigeration cycle may circulate a series of processes composed of compression-condensation-expansion-evaporation, and the refrigeration cycle may be configured to supply air that is conditioned by changing heat with a refrigerant.

(21) The compressor compresses refrigerant gas, and discharges the high temperature and high pressure refrigerant gas. The discharged refrigerant gas flows into the condenser. The condenser condenses the compressed refrigerant into a liquid phase and releases heat to the surroundings through a condensation process.

(22) The expansion valve expands the high temperature and high pressure liquid refrigerant, which is condensed by the condenser, into a low pressure liquid refrigerant. The evaporator evaporates the refrigerant, which is expanded in the expansion valve, and returns the low temperature and low pressure refrigerant gas to the compressor. The evaporator may exchange heat with an object, which is to be cooled, by using evaporation latent heat, thereby obtaining the refrigeration effect. Through this cycle, the air conditioner may adjust a temperature of an indoor space.

(23) An outdoor unit of an air conditioner refers to a part of the refrigeration cycle composed of a compressor and an outdoor heat exchanger. An indoor unit of an air conditioner includes an indoor heat exchanger, and an expansion valve may be located in either an indoor unit or an outdoor unit. An indoor heat exchanger and an outdoor heat exchanger serve as a condenser or an evaporator. When the indoor heat exchanger is used as the condenser, the air conditioner operates as a heater, and when the indoor heat exchanger is used as the evaporator, the air conditioner operates as a cooler.

(24) Hereinafter embodiments according to the disclosure will be described in detail with reference to the accompanying drawings.

(25) FIG. 1 is a perspective view of an air conditioner according to an embodiment of the disclosure, FIG. 2 is a side sectional view of the air conditioner according to an embodiment of the disclosure, and FIG. 3 is a view illustrating a state in which a panel assembly of the air conditioner of FIG. 1 is lowered.

(26) Referring to FIGS. 1 and 2, an air conditioner **1** according to an embodiment may be installed on a ceiling C. At least a part of the air conditioner **1** may be embedded in the ceiling C. The air conditioner **1** according to an embodiment of the disclosure may be provided in a 4-way type. However, the disclosure is not limited thereto, and the air conditioner **1** may be provided as an air conditioner in a 2-way type or an air condition including an outlet in an annular shape.

(27) The air conditioner **1** may include a housing **10** including an outlet **15**, a heat exchanger **20** provided in an inside of the housing **10**, and a blower fan **30** configured to allow air to flow.

(28) The housing **10** may be formed in the form of a rectangular enclosure opened downward to accommodate components of the air conditioner **1**. The housing **10** may include an upper housing **11** disposed in the inside of the ceiling C and a lower housing **13** coupled to a lower side of the upper housing **11**. The upper housing **11** may be suspended from the ceiling C without being disposed in the inside of the ceiling C.

(29) However, the disclosure is not limited thereto, and the upper housing **11** and the lower housing **13** may be integrally formed as a single housing **10**.

(30) An inlet portion **14** configured to allow air, which is sucked through a panel assembly **100**, to

flow to the blower fan **30** may be formed on a central part of the lower housing **13**. The outlet **15**, through which air is discharged, may be formed on the outside of the inlet portion **14**. An inlet flow path **P1** through which air, which is sucked through the inlet portion **14**, flows may be provided between the inlet portion **14** and the blower fan **30**. An outlet flow path **P2** through which air, which is discharged by the blower fan **30**, flows may be provided between the blower fan **30** and the outlet **15**.

(31) The outlet **15** may be formed adjacent to each edge to correspond to an outer periphery of the lower housing **13**. Four outlets **15** may be formed. That is, two outlets **15** may be formed in an X-axis direction and two outlets **15** may be formed in a Y-axis direction. The four outlets **15** may be arranged to discharge air to all four sides of the room. With this structure, the air conditioner **1** may suck air from the lower side, and cool or heat the air and then discharge the cooled or heated air to the lower side.

(32) On the bottom surface of the lower housing **13**, the panel assembly **100** including an inlet grille **112** in which a plurality of inlets **111** is formed may be arranged. Air flowing through the plurality of inlets **111** may flow toward the blower fan **30** side through the inlet portion **14**.

(33) The panel assembly **100** may be coupled to the lower housing **13**. The panel assembly **100** may be provided to cover a lower side of the lower housing **13** and attachable to or detachable from the lower side of the lower housing **13**.

(34) Particularly, the panel assembly **100** may be attachable to or detachable from the housing **10** and may be configured to be lifted and lowered in an up and down direction. When the panel assembly **100** is moved downward, the panel assembly **100** may be separated from the housing **10**, and when the panel assembly **100** is moved upward, the panel assembly **100** may be coupled to the housing **10**, again.

(35) A lifting and lowering device **200** configured to lift and lower the panel assembly **100** may be provided in the inside of the housing **10**.

(36) The panel assembly **100** may include a panel **110** including the plurality of inlets **111**, a first filter **120** configured to remove foreign substance in air, which flows through the plurality of inlets **111**, and a second filter **130** arranged above the first filter **120** and configured to secondly filter air that is filtered by the first filter **120**.

(37) The first filter **120** may be formed of a mesh material so as to filter out foreign substances. The second filter **130** may be provided as an electrostatic precipitator filter that is supplied with power so as to electrically collect foreign substance in air.

(38) The panel assembly **100** and the lifting and lowering device **200** will be described later in detail.

(39) The heat exchanger **20** may be formed in a square ring shape and arranged on the outer side of the blower fan **30** inside the housing **10**. The heat exchanger **20** is not limited to a square ring shape, and may be provided in various shapes such as a circular, elliptical, or polygonal shape.

(40) The heat exchanger **20** may be placed on a drain tray **16**, and thus condensation water generated in the heat exchanger **20** may be collected in the drain tray **16**. The drain tray **16** may be formed in a shape corresponding to the shape of the heat exchanger **20**. That is, when the heat exchanger **20** is in the form of a square ring, the drain tray **16** may also be in the form of a square ring, and alternatively, when the heat exchanger **20** is in the form of a circular shape, the drain tray **16** may also be in the form of a circular shape.

(41) The blower fan **30** may be arranged at a central portion of the housing **10**. That is, the blower fan **30** may be provided in the inner side of the heat exchanger **20**. The blower fan **30** may be a centrifugal fan configured to suck air in an axial direction and discharge the sucked air to a radial direction. The air conditioner **1** may be provided with a blowing motor **31** configured to drive the blower fan **30**.

(42) A blade **40** may be rotatably provided on the outlet **15**. The blade **40** may guide the direction of air discharged through the outlet **15**.

(43) The panel assembly **100** may be configured to be lifted and lowered, as shown in FIG. 3.

(44) The lifting and lowering device **200** may include a motor **210** (refer to FIG. 4) and a wire **220** including one end connected to the motor **210** and the other end connected to the panel **110**.

(45) The panel assembly **100** may be connected to the wire **220** and moved in the up and down direction through an operation in which the wire is wound and unwound by driving the motor **210**. That is, when the motor **210** is rotated in one direction, the panel assembly **100** may be lowered, and when the motor **210** is rotated in an opposite direction, the panel assembly **100** may be lifted.

(46) In order to maintain a room temperature and at the same time to purify indoor air, the air conditioner **1** includes the first filter **120** and the second filter **130**. Further, in order to increase the air purification efficiency, a user may need to periodically remove foreign substances collected in the first filter **120** and the second filter **130**.

(47) At this time, the air conditioner **1** is installed on the ceiling C, and thus it may be difficult to perform maintenance on the first filter **120** and the second filter **130** of the air conditioner **1**. In a conventional manner, a user may have a difficulty in which through a piece of equipment such as a ladder, the user disassembles the filter from the air conditioner, which is installed on the ceiling, to clean the filter and mount the filter to the air conditioner installed on the ceiling.

(48) In order to relieve such a difficulty, an air conditioner in the conventional manner is disclosed, and the air conditioner is configured in such a way that by lifting and lowering a panel on which a filter is arranged, the air conditioner allows a user to easily separate the filter from the panel, which is lowered to a user's height, without directly disassembling the filter from the ceiling.

(49) However, as for an air conditioner including an electrostatic precipitator filter according to an embodiment of the disclosure, the electrostatic precipitator filter is coupled to the housing rather than the panel because power needs to be supplied to the electrostatic precipitator filter, and a load of the electrostatic precipitator filter is heavy.

(50) At this time, when the panel is lowered by the lifting and lowering device, a user may easily separate the filter including the mesh material that is lowered along with the panel. However, even when the panel is lowered by the lifting and lowering device, a user still needs to separate the electrostatic precipitator filter assembled to the ceiling side through a piece of equipment such as a ladder so as to perform maintenance on the electrostatic precipitator filter because the electrostatic precipitator filter is arranged in the inside of the housing embedded in the ceiling.

(51) Particularly, the electrostatic precipitator filter is configured to collect foreign substances of a fine size and configured to have a high dust collecting rate. Therefore, in order to increase air cleaning efficiency, the electrostatic precipitator filter is required to manage foreign substances collected in the filter on a certain period shorter than that of a general filter. However, as described above, in the conventional manner, the electrostatic precipitator filter is arranged in the inside of the housing embedded in the ceiling, and thus maintenance of the electrostatic precipitator filter is more difficult than that of the general filter.

(52) To relieve this difficulty, the air conditioner **1** according to an embodiment of the disclosure includes the filter assembly **100** configured to be lifted and lowered from the ceiling C, and the filter assembly **100** includes the first filter **120** including a mesh material and the second filter **130** provided as the electrostatic precipitator filter configured to collect foreign substances in the air using electricity. Accordingly, when a user lowers the filter assembly **100**, the user can easily perform maintenance on the filter **120** including a mesh material and the second filter **130** provided as the electrostatic precipitator filter.

(53) Particularly, because the panel assembly **100** including the first filter **120** and the second filter **130** is configured to be lifted and lowered by the lifting and lowering device **200**, the panel **110**, the first filter **120** and the second filter **130** may be integrally lifted and lowered and thus a user can separate the second filter **130** as well as the first filter **120** from the air conditioner **1** and perform maintenance on the second filter **130** as well as the first filter **120**.

(54) Hereinafter the panel assembly **100** and the lifting and lowering device **200** will be described

in detail.

(55) FIG. 4 is a view illustrating a part of configuration of the air conditioner of FIG. 3, FIG. 5 is a bottom view of a housing of the air conditioner according to an embodiment of the disclosure, FIG. 6 is an exploded perspective view of the panel assembly of the air conditioner according to an embodiment of the disclosure, and FIG. 7 is a cross-sectional perspective view illustrating a part of configuration when the panel assembly of the air conditioner according to an embodiment of the disclosure is lifted.

(56) As illustrated in FIGS. 4 and 5, the air conditioner 1 may include a support panel 50 provided in a central part of the lower housing 13. The lifting and lowering device 200 may be arranged above the support panel 50.

(57) However, the disclosure is not limited thereto, and the support panel 50 may be integrally formed with the lower housing 13 as a part of the lower housing 13, and alternatively, the support panel 50 may be integrally formed with the housing 10 as a part of the housing 10.

(58) The lifting and lowering device 200 may include a plurality of motors 210. It is appropriate that the lifting and lowering device 200 may include four motors 210. The panel assembly 100 may be stably lifted and lowered because four motors 210 are provided.

(59) The wire 220 may be connected to each motor 210. Accordingly, it is appropriate that four wires 220 are provided. One end of each of the four wires 220 may be connected to each motor 210 and the other end of each of the four wires 220 may be connected to the panel 110, respectively.

(60) This is because the panel assembly 100 includes the second filter 130 having a relatively large load and thus a relatively large force is required to lift the panel assembly 100. In addition, through the four wires 220 connected to the four motors 210, the four motors 210 may lift and lower the panel assembly 100 while maintaining the balance with respect to a horizontal direction of the panel assembly 100.

(61) Two of the four wires 220 may be connected to a vicinity of a first side 115 of the panel 110 and the other two of the four wires 220 may be connected to a vicinity of a second side 116 parallel to the first side 115 of the panel 110.

(62) The two wires 220 connected to the vicinity of the first side 115 and the two wires 220 connected to the vicinity of the second side 116 may be symmetrically connected to the panel 110 with respect to the central part of the panel 110.

(63) Accordingly, when the panel assembly 100 is lifted and lowered, the lifting and lowering device 200 may lift and lower the panel assembly 100 in a balanced manner without being inclined to one side.

(64) As the panel assembly 100 is lifted and lowered by the four wires 220, the panel 110, the first filter 120, and the second filter 130 may be lifted and lowered in a state of being arranged in a vertical direction with respect to a lifting and lowering direction.

(65) That is, the panel assembly 100 may be lifted and lowered in a horizontal state, and thus the first filter 120 and the second filter 130 may also be lifted and lowered in the horizontal state. Accordingly, the panel assembly 100 may be lowered to a range in which a user can hold the panel assembly 100 and thus the user may easily separate the first filter 120 and the second filter 130 from the panel 110.

(66) This is because when the first filter 120 or the second filter 130 is lowered in an inclined state, it is difficult for the user to hold each of the filters 120 and 130 and thus it is difficult for the user to separate the filters 120 and 130 from the panel 110.

(67) In addition, because the user presses the panel 110 upon separating the filters 120 and 130, it is possible to prevent the filter assembly 100 from being severely moved in a direction in which the user presses the panel 110.

(68) The support panel 50 may include a through hole 51 provided to allow the wire 220 to penetrate and extend downward. The through hole 51 may be provided to correspond to the number of the wire 220. According to an embodiment of the disclosure, four through holes 51 may be

provided. The support panel **50** may include a guide **52** configured to guide a lifting direction of the panel assembly **100**, which is described later.

(69) The central part of the support panel **50** may include an opening **50a** provided to allow air sucked through the panel assembly **100** to flow.

(70) The second filter **130** may be inserted into the opening **50a** upon the coupling between panel assembly **100** and the housing **10**.

(71) The panel assembly **100** may include the panel **110** configured to cover the lower side of the housing **10** as described above.

(72) The panel **110** may include the inlet grille **112** forming the plurality of inlets **111**. The upper side of the panel **110** may include a frame **113** provided to allow the first filter **120** and the second filter **130** to be coupled thereto.

(73) The first filter **120** may be arranged above the inlet grille **112** and the second filter **130** may be arranged above the first filter **120**. Accordingly, the inlet grille **112**, the first filter **120** and the second filter **130** may be arranged to be stacked in the up and down direction.

(74) The second filter **130** may be separated from and coupled to the frame **113** the up and down direction corresponding to the lifting and lowering direction of the filter assembly **100**.

(75) The first filter **120** may be separated from and coupled to the frame **113** in a direction perpendicular to the up and down direction.

(76) Alternatively, each of the filters **120** and **130** may have a separating direction and a coupling direction different from each other and thus the filters **120** and **130** may be separated from and coupled to the frame **113** independently of each other without limitation in each of the separation and coupling state.

(77) The first filter **120** may include a mesh member **121**. The mesh member **121** may be arranged on the first filter **120** in a zig-zag shape in the up and down direction to maximize a surface area (refer to FIG. 2).

(78) When the air is sucked into the filter assembly **100**, foreign substances having a predetermined size may be filtered out by the mesh member **121** while passing through the first filter **120**. However, foreign substances having a size smaller than the through hole of the mesh member **121** may pass through the first filter **120**.

(79) The foreign substances having a size smaller than the through hole of the mesh member **121** may be filtered out by the second filter **130**. That is, the air sucked into the air conditioner **1** may be filtered twice while passing through the panel assembly **100** and thus the foreign substances in the air may be efficiently collected in the panel assembly **100**.

(80) The second filter **130** may be provided as an electrostatic precipitator filter as described above.

(81) The electrostatic precipitator filter may adsorb foreign substances in the charged air and electrically collect the foreign substances in the air.

(82) Accordingly, the second filter **130** may need to be supplied with high-voltage power to electrically collect foreign substances in the air.

(83) When the electrostatic precipitator filter corresponding to the second filter **130** is disposed in the inside of the housing in the conventional manner, it is possible to stably supply a high voltage to the electrostatic precipitator filter. However, the air conditioner **1** according to an embodiment of the disclosure has a difficulty in supplying a high voltage to the second filter **130** because the second filter **130** is detachable from the housing **10** and lifted and lowered in the up and down direction.

(84) To relieve this difficulty, the air conditioner according to an embodiment of the disclosure may include a power supply (not shown) configured to stably supply a voltage to the second filter **130** in response to the coupling between the panel assembly **100** and the housing **10**.

(85) As illustrated in FIGS. 5 to 7, the power supply (not shown) may include an intermediate member **70** disposed in an inside **10a** of the housing **10** and configured to supply a voltage to the second filter **130** in response to the coupling between the panel assembly **100** and the housing **10**.

Particularly, the intermediate member **70** may be arranged in the inside of the opening **50a** of the support panel **50**.

(86) However, the disclosure is not limited thereto, and the intermediate member **70** may be disposed on the support panel **50**.

(87) Although not shown in the drawing, a power supply (not shown) may be also disposed in the inside **10a** of the housing **10** and configured to supply a high voltage to the second filter **130** through the intermediate member **70** by receiving power supplied from the outside. In addition, the power supply (not shown) may be disposed above the support panel **50**.

(88) The intermediate member **70** may be electrically connected to the second filter **130** and supply the power to the second filter **130** when the panel assembly **100** is lifted and the panel assembly **100** and the housing **10** are coupled to each other.

(89) When the panel assembly **100** is lowered, the electrical connection between the intermediate member **70** and the second filter **130** may be released and the supply of power to the second filter **130** may be blocked.

(90) The second filter **130** may include a contact portion **131** configured to be in contact with the intermediate member **70** electrically in response to the coupling between the panel assembly **100** and the housing **10**. As described above, the intermediate member **70** is disposed in the inside of the opening **50a** of the support panel **50**, and the second filter **130** may be inserted into the inside of the opening **50a** of the support panel **50** in response to the coupling between the panel assembly **100** and the housing **10**. Accordingly, the intermediate member **70** may be in contact with the contact portion **131** in response to the coupling between the panel assembly **100** and the housing **10**.

(91) In response to the coupling between the panel assembly **100** and the housing **10**, the contact portion **131** may be lifted to a position in which the contact portion **131** is in contact with the intermediate member **70** by pressing the intermediate member **70**. The contact portion **131** and the intermediate member **70** may be disposed on the same line in the up and down direction.

(92) The contact portion **131** is a portion in which a contact point is formed by being in physical contact with the intermediate member **70**.

(93) The contact portion **131** may be supplied with a voltage from the intermediate member **70** upon being in contact with the intermediate member **70**, and when the contact point is shorted upon the release of the contact between the contact portion **131** and the intermediate member **70**, the supply of voltage may be blocked.

(94) The contact portion **131** may be disposed toward the lifting and lowering direction of the panel assembly **100**. Accordingly, the contact portion **131** may be in contact with the intermediate member **70** in the lifting direction of the panel assembly **100**.

(95) It is appropriate that the contact portion **131** is disposed on an upper surface of the second filter **130**. Therefore, upon the lifting of the second filter **130**, the second filter **130** may be in contact with the intermediate member **70** by pressing the intermediate member **70** in the lifting direction. Upon the lowering of the second filter **130**, the pressure applied to the intermediate member **70** may be released and the contact with the intermediate member **70** may be released.

(96) The intermediate member **70** may be provided in a plate spring shape. Accordingly, when the contact portion **131** is lifted and in contact with the intermediate member **70**, the contact portion **131** may press the intermediate member **70** upward and the intermediate member **70** may be biased downward. Therefore, the intermediate member **70** may be maintained at a state of being in constant contact with the contact portion **131**.

(97) That is, the contact point formed on the second filter **130** may be formed in the lifting and lowering direction or the up and down direction. Accordingly, the contact point with the intermediate member **70** may be stably formed even when the second filter **130** is lifted and lowered in the up and down direction. As a result, the second filter **130** may be electrically connected to the power supply (not shown) stably.

(98) Therefore, even when the second filter **130** is movable with the panel **110**, the second filter **130** may be stably supplied with the voltage through the contact point formed in a moving direction. Therefore, when the second filter **130** is lowered to a range in which a user generally holds and then the second filter **130** is lifted and arranged in the inside of the housing **30**, the contact point with the intermediate member **70** may be stably formed and thus the second filter **130** may be stably supplied with the power.

(99) The power supply (not shown) may include a power supply device configured to supply power and a high voltage device configured to generate a high voltage using the supplied power and transmit the high voltage to the intermediate member **70**.

(100) At this time, as for the air conditioner **1** according to an embodiment of the disclosure, both the power supply device and the high voltage device of the power supply (not shown) may be installed on the upper side of the support panel **50** or in the inside **10a** of the housing **10**. The high voltage device may be electrically connected to the second panel **130** through the intermediate member **70**.

(101) However, the disclosure is not limited thereto, and the high voltage device among the power supply (not shown) may be disposed in the panel assembly **100** and lifted and lowered together with the panel assembly **100**.

(102) At this time, the high voltage device disposed in the panel assembly **100** may be maintained in the electrical connection with the second filter **130**, and when the second filter **130** is inserted into the inside **10a** of the housing **10**, the second filter **130** may be electrically connected to the power supply device through the intermediate member **70**.

(103) That is, a part of the power supply (not shown) may be disposed in the panel assembly **100** and then lifted and lowered along with the panel assembly **100**. The supply of power to some components of the power supply (not shown) may be blocked upon the lowering, and when the panel assembly **100** is lifted and then coupled to the housing **10**, the some components of the power supply (not shown) may be in contact with the intermediate member **70** arranged in the housing **10**, thereby receiving the power.

(104) Hereinafter the guide **52** configured to guide the lifting direction of the panel assembly **100** will be described in detail.

(105) FIG. **7** is a cross-sectional perspective view illustrating a part of configuration when the panel assembly of the air conditioner according to an embodiment of the disclosure is lifted, and FIG. **8** is a cross-sectional view illustrating a part of configuration when the panel assembly of the air conditioner according to an embodiment of the disclosure is lifted.

(106) As mentioned above, the panel assembly **100** may be lifted and lowered according to the operation of the wire **220** that is wound and unwound. At this time, as the wire **220** is wound, the panel assembly **100** may be shaken in the process of lifting or the panel assembly **100** may be shaken by an external force.

(107) When the panel assembly **100** is not coupled to the housing **10** in a correct position in the up and down direction due to the shaking, the contact portion **131** may be affected by the shaking and thus the contact portion **131** may not press the intermediate member **70** in a correct position with respect to the up and down direction.

(108) Accordingly, the contact portion **131** and the intermediate member **70** may not be electrically connected to each other, or a short circuit may occur. To prevent this, the air conditioner **1** according to an embodiment of the disclosure may include the guide **52** configured to guide the lifting direction of the panel assembly **100**.

(109) The guide **52** may guide the lifting direction of the panel assembly **100** to accurately move the contact portion **131** to a position in which the intermediate member **70** is arranged.

(110) The guide **52** may be disposed on the support panel **50**. Particularly, the guide **52** may be formed along the opening **50a** of the support panel **50**.

(111) The guide **52** may include a guide surface **52a** formed in a convex shape in the direction of

the opening **50a**. The guide surface **52a** may be formed as a curved surface. The guide **52** may be formed on four inner edges of the support panel **50** forming the opening **50a**, respectively.

(112) In addition, the guide **52** may also be formed on four corners that is formed when the four inner edges of the support panel **50** are in contact to each other.

(113) However, the disclosure is not limited thereto, and the guide **52** may be formed on at least two of the four inner edges and may be formed on at least two of the four corners.

(114) As mentioned above, when the second filter **130** is inserted into the inside of the opening **50a**, the panel assembly **100** may be coupled to the housing **10**. The guide **52** formed along the opening **50a** may guide the lifting direction of the second filter **130** and the frame **113** of the panel **110** configured to fix the second filter **130** to the panel **110** and thus the second filter **130** may be stably inserted into the opening **50a**.

(115) That is, when the panel assembly **100** is lifted while being shaken, the second filter **130** may be disposed on the outside of the opening **50a** in the up and down direction, but the side surface of the second filter **130** may come into contact with the guide surface **52a** and the lifting direction may be guided to the inside of the opening **50a**.

(116) In addition, when the second filter **130** is lifted while being shaken in the inside of the opening **50a**, the side surface of the frame **113** may come into contact with the guide surface **52a** so as to guide the lifting direction of the panel assembly **100**, thereby allowing the second filter **130** to be lifted in the vertical direction.

(117) The frame **113** may include a guide portion **114** provided to allow the panel assembly **100** to be stably guided along the guide surface **52a** upon being in contact with the guide surface **52a**.

(118) The guide portion **114** may be provided in a tapered shape including a cross section thereof being reduced toward the upper side, or may be formed in a curved surface being concave toward the upper side. Therefore, when the guide portion **114** is in contact with the guide surface **52a**, the guide portion **114** may be stably moved along the guide surface **52a** and thus the lifting direction of the panel assembly **100** may be guided.

(119) As described above, the contact portion **131** may be lifted while being arranged on the substantially same line as the intermediate member **70** in the up and down direction by the guide **52**. The contact portion **131** may be stably in contact with the intermediate member **70** at a correct position, thereby forming the contact point.

(120) Hereinafter a connection structure of the wire **220** and the panel assembly **100** will be described in detail.

(121) FIG. **9** is an enlarged view illustrating a part of configuration of the panel assembly of the air conditioner according to an embodiment of the disclosure.

(122) The wire **220** may be connected to the upper surface of the panel **110**.

(123) The panel **110** may include an elastic member **119** including one end connected to the other end of the wire **220** and the other end connected to the upper surface of the panel **110**.

(124) As mentioned above, the four wires **220** are provided, and accordingly, the four elastic members **119** may be provided to be connected to the other end of each wire **220**.

(125) The panel assembly **100** is supported by the four wires **220**, and thus when the tension is not equally generated on the four wires **220**, the panel assembly **100** may be lifted and lowered in an inclined state.

(126) That is, when the panel assembly **100** is inclined to one side due to an external force generated during the lifting and lowering, or when a difference in a length of the wires **220** is generated due to a difference in a degree of winding or unwinding of each wire **220**, the same tension may not be generated on the four wires **220**.

(127) Accordingly, the elastic member **119** may be coupled to the other end of each wire **220**, and a magnitude of tension which may be different on the four wires **220** may be offset by the elastic force of the elastic member **119**.

(128) In addition, even when the difference in the length of the four wires **220** is generated, the

elastic member **119** may be extended and contracted so as to offset the difference in the length of the four wires **220**, thereby stably lifting and lowering the panel assembly **100**.

(129) The elastic member **119** may be provided as a coil spring. The elastic member **119** may be extended and biased in a direction perpendicular to the lifting and lowering direction of the panel assembly **100**, thereby transmitting the elastic force in the direction perpendicular to the lifting and lowering direction of the panel assembly **100**.

(130) The panel **110** may further include a pulley **118** through which the other end of the wire **220** passes and configured to change an extending direction of the wire **220** into the horizontal direction with respect to the extending direction of the panel assembly **100**.

(131) The pulley **118** may be provided to correspond to the number of wires **220**. Therefore, the air conditioner **1** according to an embodiment of the disclosure may include four pulleys **118**.

(132) The pulley **118** may be disposed on the upper surface of the panel **110**. As described above, one end of the wire **220** may be connected to the motor **210** disposed in the inside **10a** of the housing **10** and the other end of the wire **220** may be extended in the downward direction corresponding to the lowering direction of the panel assembly **100**.

(133) By the pulley **118**, the extending direction of the other end of the wire **220** may be changed to the left and right direction or the front and rear direction perpendicular to the downward direction. That is, the other end of the wire **220** may be extended in the lifting and lowering direction of the panel assembly **100**, and changed to a direction perpendicular to the lifting and lowering direction of the panel assembly **110** by the pulley **118**.

(134) Accordingly, the other end of the wire **220** may be connected to the elastic member **119**, which is to be biased in the direction perpendicular to the lifting and lowering direction of the panel assembly **110**, in a direction in which the elastic member **119** is biased.

(135) The elastic member **119** may be extended and contracted in the direction perpendicular to the lifting and lowering direction of the panel assembly **100** when the difference in the tension or the difference in length is generated in each wire **220**.

(136) At this time, when the wire **220** is directly connected to the elastic member **119** without passing through the pulley **118**, an elastic length of each elastic member **119** may be different from each other. Accordingly, the wire **220** may not horizontally lift and lower the panel assembly **100** but the wire **220** may lift and lower the panel assembly **100** in a state in which one side of the panel assembly **100** is inclined.

(137) This is because a point in which the wire **220** and the panel **110** are connected is changed with respect to the direction perpendicular to the lifting and lowering direction of the panel assembly **100**, according to the elastic length of the elastic member **119**.

(138) When it is assumed that a point in which the wire **220** is connected to the panel **110** in a constant state of the elastic member **119** is a point in which the panel assembly **100** is horizontally lifted and lowered, a point in which the wire **220** is connected to the panel **110** may be changed according to an extended state or a contracted state of the elastic member **119** and thus the panel assembly **100** may not be horizontally lifted and lowered.

(139) However, through the pulley **118**, it is possible to horizontally lift and lower the panel assembly **100** regardless of the elastic length of each elastic member **119**. This is because, by the pulley **118**, the wire **220** is connected to the panel **110** only at a point, in which the pulley **118** is arranged in the direction perpendicular to the lifting and lowering direction of the panel assembly **100**, regardless of the elastic length of the elastic member **119**.

(140) In addition, by the pulley **118**, the extending direction of the wire **220** may be guided from the lifting and lowering direction of the panel assembly **100** to the direction perpendicular to the lifting and lowering direction of the panel assembly **100**. Therefore, the panel assembly **100** may be lifted and lowered while maintaining a level with respect to the ground.

(141) The panel assembly **100** may be lifted and lowered while maintaining balance through the four wires **220**. The panel assembly **100** may be lifted and lowered in the horizontal state through

the four wires **220**.

(142) In addition, even when the external force is applied during the lifting and lowering of the panel assembly **100**, the panel assembly **100** may be lifted and lowered while maintaining balance, through the four wires **220**, the four elastic members **119** connected to the four wires **220**, and the four pulleys **118** through which the four wires **220** passes.

(143) Therefore, the panel **110**, the first filter **120**, and the second filter **130** may be always lowered to the range, in which a user can hold the panel assembly **100**, in the horizontal state. Accordingly, a user can easily separate the first filter **120** and the second filter **130** from the panel **110** and easily couple the first filter **120** and the second filter **130** to the panel **110** again.

(144) Hereinafter an intermediate member **80** of an air conditioner **1** according to another embodiment of the disclosure will be described. A configuration other than the intermediate member **80** described below is the same as the above-mentioned configuration of the air conditioner **1** according to an embodiment, and thus the description thereof will be omitted.

(145) FIG. **10** is a cross-sectional perspective view illustrating a part of configuration when a panel assembly of an air conditioner according to another embodiment of the disclosure is lifted.

(146) The intermediate member **80** may be provided in a coil spring shape. The intermediate member **80** may be provided to extend in a direction corresponding to a lifting and lowering direction of a panel assembly **100**.

(147) Accordingly, when a contact portion **131** is lifted and then comes into contact with the intermediate member **80**, the contact portion **131** may press the intermediate member **80** upward and the intermediate member **80** may be biased downward so as to be in constant contact with the contact portion **131**.

(148) The contact portion **131** may press the intermediate member **80** while being moved upward, and thus the intermediate member **80** may be biased downward while being compressed upward.

(149) Hereinafter a guide **53** of the air conditioner **1** according to another embodiment of the disclosure will be described. A configuration other than the guide **53** described below is the same as the above-mentioned configuration of the air conditioner **1** according to an embodiment, and thus the description thereof will be omitted.

(150) FIG. **11** is a bottom view illustrating a support panel of a housing of an air conditioner according to still another embodiment of the disclosure, FIG. **12** is a cross-sectional view illustrating a part of configuration when a panel assembly of the air conditioner according to still another embodiment of the disclosure is lifted, and FIG. **13** is a cross-sectional view illustrating a part of configuration when the panel assembly of the air conditioner according to still another embodiment of the disclosure is lifted.

(151) As illustrated in FIGS. **11** and **12**, the guide **53** may include a guide rib **53a** formed to be convex toward a direction of an opening **50a**. The guide rib **53a** may be formed in a curve. The guide ribs **53a** may be formed in plural on each of the four inner edges of a support panel **50** forming the opening **50a**.

(152) That is, the guide rib **53a** may be provided to protrude toward a center direction of the opening **50a** from an inner wall **50b** of the support panel **50** forming the opening **50a**.

(153) When the panel assembly **100** is lifted while being shaken, the second filter **130** may be disposed on the outside of the opening **50a** in the up and down direction, but the side surface of the second filter **130** may come into contact with the guide rib **53a** and the lifting direction may be guided to the inside of the opening **50a**.

(154) In addition, when the second filter **130** is lifted while being shaken in the inside of the opening **50a**, the side surface of the frame **113** may come into contact with the guide rib **53a** so as to guide the lifting direction of the panel assembly **100**, thereby allowing the second filter **130** to be lifted in the vertical direction.

(155) In addition, as illustrated in FIG. **13**, the guide **53** may include a guide roller **53b** arranged on the inner wall **50b** of the support panel **50**.

(156) The guide roller **53b** may be provided as a pair on the inner wall **50b** including at least two inner walls that is parallel to each other.

(157) When the panel assembly **100** is lifted while being shaken, the second filter **130** may be disposed on the outside of the opening **50a** in the up and down direction, but the side surface of the second filter **130** may come into contact with the guide roller **53b** and the guide roller **53b** may be rolled by the friction force.

(158) The guide roller **53b** may guide the second filter **130** to be lifted in a tangential direction of the roller of the guide roller **53b** through rolling.

(159) In addition, when the second filter **130** is lifted while being shaken in the inside of the opening **50a**, the side surface of the frame **113** may come into contact with the guide roller **53b** so as to guide the lifting direction of the panel assembly **100**, thereby allowing the second filter **130** to be lifted in the vertical direction.

(160) As is apparent from the above description, the air conditioner may include the electrostatic precipitator filter configured to be lifted and lowered along with the panel so as to allow a user to easily perform on maintenance on the electrostatic precipitator filter.

(161) Further, because the electrostatic precipitator filter includes the contact portion configured to be in vertical contact with the power supply arranged on the housing, it is possible to easily supply power to the electrostatic precipitator filter when the panel and the electrostatic precipitator filter are lifted and then the panel is coupled to the housing.

(162) Although a few embodiments of the disclosure have been shown and described, it would be appreciated by those skilled in the art that changes may be made in these embodiments without departing from the principles and spirit of the disclosure, the scope of which is defined in the claims and their equivalents.

Claims

1. An air conditioner comprising: a housing comprising an outlet; a panel attachable to or detachable from a lower side of the housing, the panel comprising an inlet to introduce air therethrough; a first filter installable to the panel to filter the introduced air; a second filter installable to the panel at an upper side of the first filter and configured to further filter the introduced air; and a lifter configured to detach the panel from the lower side of the housing by lowering the panel from an initial position to a maintenance position, and to attach the panel to the lower side of the housing by lifting the panel from the maintenance position to the initial position, wherein in response to lowering the panel to the maintenance position by the lifter, one or more of the first filter and the second filter are accessible for the maintenance, wherein a lifting and lowering direction of the panel is parallel to a vertical direction, wherein the lifter comprises: a motor; and four wires configured to lift or lower the panel, the first filter and the second filter by driving force of the motor, each of the four wires comprising a first end connected to the motor and a second end connected to the panel, wherein the first filter is configured to be detached from and installed to the panel in a first direction, wherein the second filter is configured to be detached from and installed to the panel in a second direction different from the first direction, wherein the first filter is detachable from the panel in a state in which the second filter is installed to the panel, and wherein the first filter is installable to the panel in the state in which the second filter is installed to the panel.
2. The air conditioner of claim 1, wherein the panel, the first filter and the second filter installed in the panel, is lifted or lowered in the vertical direction while maintaining a horizontal state.
3. The air conditioner of claim 1, wherein the panel further comprises a frame configured to support the first filter and the second filter against the panel, wherein the first filter is configured to be detached from and installed to the frame in the first direction, and wherein the second filter is configured to be detached from and installed to the frame in the second direction.

4. The air conditioner of claim 3, wherein the frame is formed at an upper side of the panel.
 5. The air conditioner of claim 1, wherein the motor comprises four motors, and wherein each of the four wires is connected to a respective motor of the four motors.
 6. The air conditioner of claim 1, wherein the panel comprises an elastic member comprising one end connected to the second end of the wire and another end connected to the panel.
 7. The air conditioner of claim 6, wherein the panel further comprises a pulley through which the wire passes, and wherein the second end of the wire passes through the pulley and then is arranged perpendicular to the first end of the wire.
 8. The air conditioner of claim 1, wherein the panel further comprises a guide configured to guide at least a part of the panel to move along the guide upon being in contact with the guide during the lifting of the panel so that a lifting direction of the panel is guided to be in the initial position.
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